

UNION, UNION ALL, INTERSECT and EXCEPT Questions

Please complete the Questions using tables created below

-- Employee table in Company A

```
CREATE TABLE Employees_A (  
    EmpID INT,  
    EmpName VARCHAR(50),  
    Department VARCHAR(50),  
    Salary INT  
);
```

INSERT INTO Employees_A (EmpID, EmpName, Department, Salary) VALUES

```
(1, 'Alice', 'HR', 50000),  
(2, 'Bob', 'Finance', 60000),  
(3, 'Charlie', 'IT', 70000),  
(4, 'David', 'IT', 65000),  
(5, 'Emma', 'Marketing', 55000),  
(6, 'Frank', 'Finance', 62000),  
(7, 'Grace', 'HR', 52000),  
(8, 'Hannah', 'Sales', 58000),  
(9, 'Ivy', 'Finance', 61000),  
(10, 'Jack', 'IT', 72000),  
(11, 'Employee11', 'Finance', 59000),  
(12, 'Employee12', 'HR', 51000),  
(13, 'Employee13', 'IT', 71000),  
(14, 'Employee14', 'Sales', 56000),  
(15, 'Employee15', 'Marketing', 53000),  
(16, 'Employee16', 'Finance', 64000),  
(17, 'Employee17', 'IT', 75000),  
(18, 'Employee18', 'HR', 50500),  
(19, 'Employee19', 'Sales', 60000),  
(20, 'Employee20', 'Marketing', 57000),  
(21, 'Employee21', 'Finance', 65000),  
(22, 'Employee22', 'HR', 52000),  
(23, 'Employee23', 'IT', 73500),  
(24, 'Employee24', 'Sales', 58500),  
(25, 'Employee25', 'Marketing', 55500),  
(26, 'Employee26', 'Finance', 61500),  
(27, 'Employee27', 'HR', 52500),  
(28, 'Employee28', 'IT', 74500),  
(29, 'Employee29', 'Sales', 57000),  
(30, 'Employee30', 'Marketing', 56000),
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(31, 'Employee31', 'Finance', 60500),
(32, 'Employee32', 'HR', 53000),
(33, 'Employee33', 'IT', 73000),
(34, 'Employee34', 'Sales', 57500),
(35, 'Employee35', 'Marketing', 56500),
(36, 'Employee36', 'Finance', 62500),
(37, 'Employee37', 'HR', 51500),
(38, 'Employee38', 'IT', 74000),
(39, 'Employee39', 'Sales', 59500),
(40, 'Employee40', 'Marketing', 58000),
(41, 'Employee41', 'Finance', 63500),
(42, 'Employee42', 'HR', 53500),
(43, 'Employee43', 'IT', 75500),
(44, 'Employee44', 'Sales', 58500),
(45, 'Employee45', 'Marketing', 54500),
(46, 'Employee46', 'Finance', 64500),
(47, 'Employee47', 'HR', 52500),
(48, 'Employee48', 'IT', 76500),
(49, 'Employee49', 'Sales', 60500),
(50, 'Employee50', 'Marketing', 59000),
(51, 'Employee51', 'Finance', 65500),
(52, 'Employee52', 'HR', 54000),
(53, 'Employee53', 'IT', 77500),
(54, 'Employee54', 'Sales', 61500),
(55, 'Employee55', 'Marketing', 59500),
(56, 'Employee56', 'Finance', 66500),
(57, 'Employee57', 'HR', 54500),
(58, 'Employee58', 'IT', 78500),
(59, 'Employee59', 'Sales', 62500),
(60, 'Employee60', 'Marketing', 60000),
(61, 'Employee61', 'Finance', 67500),
(62, 'Employee62', 'HR', 55000),
(63, 'Employee63', 'IT', 79500),
(64, 'Employee64', 'Sales', 63500),
(65, 'Employee65', 'Marketing', 60500),
(66, 'Employee66', 'Finance', 68500),
(67, 'Employee67', 'HR', 55500),
(68, 'Employee68', 'IT', 80500),
(69, 'Employee69', 'Sales', 64500),
(70, 'Employee70', 'Marketing', 61000),
(71, 'Employee71', 'Finance', 69500),
(72, 'Employee72', 'HR', 56000),
(73, 'Employee73', 'IT', 81500),
(74, 'Employee74', 'Sales', 65500),

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(75, 'Employee75', 'Marketing', 61500),
(76, 'Employee76', 'Finance', 70500),
(77, 'Employee77', 'HR', 56500),
(78, 'Employee78', 'IT', 82500),
(79, 'Employee79', 'Sales', 66500),
(80, 'Employee80', 'Marketing', 62000),
(81, 'Employee81', 'Finance', 71500),
(82, 'Employee82', 'HR', 57000),
(83, 'Employee83', 'IT', 83500),
(84, 'Employee84', 'Sales', 67500),
(85, 'Employee85', 'Marketing', 62500),
(86, 'Employee86', 'Finance', 72500),
(87, 'Employee87', 'HR', 57500),
(88, 'Employee88', 'IT', 84500),
(89, 'Employee89', 'Sales', 68500),
(90, 'Employee90', 'Marketing', 63000),
(91, 'Employee91', 'Finance', 73500),
(92, 'Employee92', 'HR', 58000),
(93, 'Employee93', 'IT', 85500),
(94, 'Employee94', 'Sales', 69500),
(95, 'Employee95', 'Marketing', 63500),
(96, 'Employee96', 'Finance', 74500),
(97, 'Employee97', 'HR', 58500),
(98, 'Employee98', 'IT', 86500),
(99, 'Employee99', 'Sales', 70500),
(100, 'Employee100', 'Marketing', 64000);
```

-- Employee table in Company B

```
CREATE TABLE Employees_B (
    EmpID INT,
    EmpName VARCHAR(50),
    Department VARCHAR(50),
    Salary INT
);
```

INSERT INTO Employees_B (EmpID, EmpName, Department, Salary) VALUES

```
(3, 'Charlie', 'IT', 70000),
(4, 'David', 'IT', 64000),
(6, 'Frank', 'Finance', 62000),
(7, 'Grace', 'HR', 52000),
(8, 'Hannah', 'Sales', 58000),
(101, 'Employee101', 'Finance', 63000),
(102, 'Employee102', 'HR', 54000),
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(103, 'Employee103', 'IT', 76000),
(104, 'Employee104', 'Sales', 59000),
(105, 'Employee105', 'Marketing', 58000),
(106, 'Employee106', 'Finance', 60000),
(107, 'Employee107', 'HR', 51000),
(108, 'Employee108', 'IT', 72500),
(109, 'Employee109', 'Sales', 56500),
(110, 'Employee110', 'Marketing', 57500),
(111, 'Employee111', 'Finance', 63500),
(112, 'Employee112', 'HR', 52000),
(113, 'Employee113', 'IT', 74500),
(114, 'Employee114', 'Sales', 58500),
(115, 'Employee115', 'Marketing', 55500),
(116, 'Employee116', 'Finance', 61500),
(117, 'Employee117', 'HR', 52500),
(118, 'Employee118', 'IT', 75500),
(119, 'Employee119', 'Sales', 59500),
(120, 'Employee120', 'Marketing', 56500),
(121, 'Employee121', 'Finance', 62500),
(122, 'Employee122', 'HR', 53000),
(123, 'Employee123', 'IT', 76500),
(124, 'Employee124', 'Sales', 60500),
(125, 'Employee125', 'Marketing', 57500),
(126, 'Employee126', 'Finance', 63500),
(127, 'Employee127', 'HR', 53500),
(128, 'Employee128', 'IT', 77500),
(129, 'Employee129', 'Sales', 61500),
(130, 'Employee130', 'Marketing', 58500),
(131, 'Employee131', 'Finance', 64500),
(132, 'Employee132', 'HR', 54000),
(133, 'Employee133', 'IT', 78500),
(134, 'Employee134', 'Sales', 62500),
(135, 'Employee135', 'Marketing', 59500),
(136, 'Employee136', 'Finance', 65500),
(137, 'Employee137', 'HR', 54500),
(138, 'Employee138', 'IT', 79500),
(139, 'Employee139', 'Sales', 63500),
(140, 'Employee140', 'Marketing', 60500),
(141, 'Employee141', 'Finance', 66500),
(142, 'Employee142', 'HR', 55000),
(143, 'Employee143', 'IT', 80500),
(144, 'Employee144', 'Sales', 64500),
(145, 'Employee145', 'Marketing', 61500),
(146, 'Employee146', 'Finance', 67500),

(147, 'Employee147', 'HR', 55500),
(148, 'Employee148', 'IT', 81500),
(149, 'Employee149', 'Sales', 65500),
(150, 'Employee150', 'Marketing', 62500),
(151, 'Employee151', 'Finance', 68500),
(152, 'Employee152', 'HR', 56000),
(153, 'Employee153', 'IT', 82500),
(154, 'Employee154', 'Sales', 66500),
(155, 'Employee155', 'Marketing', 63500),
(156, 'Employee156', 'Finance', 69500),
(157, 'Employee157', 'HR', 56500),
(158, 'Employee158', 'IT', 83500),
(159, 'Employee159', 'Sales', 67500),
(160, 'Employee160', 'Marketing', 64500),
(161, 'Employee161', 'Finance', 70500),
(162, 'Employee162', 'HR', 57000),
(163, 'Employee163', 'IT', 84500),
(164, 'Employee164', 'Sales', 68500),
(165, 'Employee165', 'Marketing', 65500),
(166, 'Employee166', 'Finance', 71500),
(167, 'Employee167', 'HR', 57500),
(168, 'Employee168', 'IT', 85500),
(169, 'Employee169', 'Sales', 69500),
(170, 'Employee170', 'Marketing', 66500),
(171, 'Employee171', 'Finance', 72500),
(172, 'Employee172', 'HR', 58000),
(173, 'Employee173', 'IT', 86500),
(174, 'Employee174', 'Sales', 70500),
(175, 'Employee175', 'Marketing', 67500),
(176, 'Employee176', 'Finance', 73500),
(177, 'Employee177', 'HR', 58500),
(178, 'Employee178', 'IT', 87500),
(179, 'Employee179', 'Sales', 71500),
(180, 'Employee180', 'Marketing', 68500),
(181, 'Employee181', 'Finance', 74500),
(182, 'Employee182', 'HR', 59000),
(183, 'Employee183', 'IT', 88500),
(184, 'Employee184', 'Sales', 72500),
(185, 'Employee185', 'Marketing', 69500),
(186, 'Employee186', 'Finance', 75500),
(187, 'Employee187', 'HR', 59500),
(188, 'Employee188', 'IT', 89500),
(189, 'Employee189', 'Sales', 73500),
(190, 'Employee190', 'Marketing', 70500),

(191, 'Employee191', 'Finance', 76500),
(192, 'Employee192', 'HR', 60000),
(193, 'Employee193', 'IT', 90500),
(194, 'Employee194', 'Sales', 74500),
(195, 'Employee195', 'Marketing', 71500),
(196, 'Employee196', 'Finance', 77500),
(197, 'Employee197', 'HR', 60500),
(198, 'Employee198', 'IT', 91500),
(199, 'Employee199', 'Sales', 75500),
(200, 'Employee200', 'Marketing', 72500);

Please Do The Question Below Using Above Tables

♦ UNION (removes duplicates)

1. List all unique employee names from **both companies**.
2. Get all unique departments from both companies.
3. Show all unique salaries from both tables.
4. Get all unique (**EmpID**, **EmpName**) combinations from both tables.
5. List all unique employees who work either in **IT** or **Finance** across both companies.

♦ UNION ALL (keeps duplicates)

1. List all employee names from both companies, including duplicates.
2. Show all department names (with duplicates) from both companies.
3. List all (**EmpID**, **EmpName**, **Salary**) from both companies including duplicates.
4. Count how many total employees (with duplicates) exist in both companies.
5. Show all employee names where duplicates should also appear (use UNION ALL).

♦ INTERSECT (common rows)

1. Find employees who exist in **both companies with exactly same data** (EmpID, EmpName, Dept, Salary).
2. Get all employee names that are present in both companies.
3. Find the departments that exist in both companies.
4. Find salaries that are common in both companies.
5. Get all employees who are in **Company A and Company B** with same EmpID and Department.

♦ EXCEPT (difference between sets)

1. Find employees who are in **Company A** but not in **Company B**.
2. Find employees who are in **Company B** but not in **Company A**.
3. Get departments that exist in **Company A but not in B**.
4. Find employees who are only in **Company A's IT department** but not in Company B's IT department.
5. Get employees who have the **same EmpID** in both companies but different salary (hint: use EXCEPT with EmpID+Salary check).